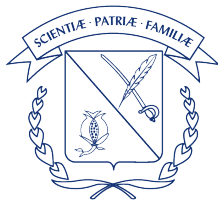


Financial Market II

Luis Francisco Gómez López

FAEDIS

2025-02-27



UNIVERSIDAD MILITAR
NUEVA GRANADA

Table of contents I

- 1 Please Read Me
- 2 Purpose
- 3 Principal instruments “*Mercado no intermediado*”
- 4 Banking market or “*Mercado Intermediado*”
- 5 Acknowledgments

- Check the message **Welcome greeting** published in the News Bulletin Board.
- Dear student please edit your profile uploading a photo where your face is clearly visible.
- The purpose of the virtual meetings is to answer questions and not to make a summary of the study material.
- If you want to participate, please fill out the following survey: Primer corte 30% > Learning Activities > Tu opinión sobre la economía colombiana
- This presentation is based on (Cardenas 2020, chap. 8)

Analyze the functioning of the financial market, identifying the types of intermediaries and instruments that are part of it

- **Fixed income (Renta fija):** provides returns in the form of regular interest payments and repayments of the principal
 - Títulos de tesorería (TES)
 - Debt securities issued by the national government and administered by the Banco de la República.
 - The national government use this instrument to finance its activities
 - Certificados de Depósito a Término (CDT)



Figure 1: Interest rates TES zero coupon (COP, 1, 5 and 10 years)

- TES zero coupon

- Actual bond price: P
- Nominal value: NV where $P < NV$
- Term: n
- Interest rate: i

$$P = \frac{NV}{(1+i)^n}$$

$$(1+i)^n = \frac{NV}{P}$$

$$i = \left(\frac{NV}{P} \right)^{\frac{1}{n}} - 1$$

- An example: TES zero coupon

- Actual bond price: $P = 600000$
- Nominal value: $NV = 1000000$
- Term: $n = 10$

$$\begin{aligned} i &= \left(\frac{NV}{P} \right)^{\frac{1}{n}} - 1 \\ &= \left(\frac{1000000}{600000} \right)^{\frac{1}{10}} - 1 \\ &\approx 0.0524 \text{ (5.24 \%)} \end{aligned}$$

• Equity (Renta variable):

• Shares/Stocks

- These instruments are issued by companies to raise funds from the general public
- They represent a fractional ownership in the company that issue them
- If the company does well, its value may increase, and the value of your shares may also rise. You can then sell your shares for a profit. If the company performs poorly, the value of your shares may decrease, and you could lose money.
- Some companies share their profits with shareholders by paying dividends.
- The decision to pay dividends or reinvest the profit rests with the company's board of directors. Whether a company pays dividends or reinvests its profits depends on its specific circumstances and strategic goals.



Figure 2: MSCI COLCAP Index

- **What is a bank?**

- “A bank is an institution **whose current operations** consist in granting loans and receiving deposits from the public” (Freixas and Rochet 2008, 1)
 - Therefore the core activities of banks are related to deposits and loans

- **What functions banks perform?** (Freixas and Rochet 2008, 2)

- Offering liquidity and payment services
- Transforming assets
- Managing risks
- Processing information and monitoring borrowers

- **Offering liquidity and payment services** (Freixas and Rochet 2008, 2–4)
 - Banks offer short term credits to companies and individuals and have created networks that facilitate the transfer of funds between the bank accounts of economic agents
- **Transforming assets** (Freixas and Rochet 2008, 5–6)
 - **Convenience of denomination:** banks collect small deposits to offer large loans
 - **Quality transformation:** bank deposits offer better risk-return characteristics than direct investments
 - **Maturity transformation:** banks transform securities with short maturities, offered to depositors, into securities with long maturities, which borrowers desire

- **Managing risks:**

- **Credit risk:** it is related to the the probability that a loan is no repaid
 - **Interest rate risk:** it is related to the difference between deposit rates, which change more, and lending rates, which are more stable
 - **Liquidity risk:** it is related to the difficulty a bank has in selling a loan compared to the ease with which a depositor withdraws his savings
- In the case of the Colombian context the framework to manage risks is known as **Sistema Integral de Administración de Riesgos (SIAR)** (Colombia 2021, chap. 31)

Table 1: Supervised banking establishments

Type	Code	Abbreviate Name	NIT
1	1	Banco de Bogotá S.A.	860002964-4
1	2	Banco Popular	860007738-9
1	6	Itaú; Banco Itaú.	890903937-0
1	7	Bancolombia	890903938-8
1	9	Citibank	860051135-4
1	12	Banco GNB Sudameris	860050750-1
1	13	BBVA Colombia	860003020-1
1	23	Banco de Occidente	890300279-4
1	30	Banco Caja Social S.A.	860007335-4
1	39	Banco Davivienda	860034313-7
1	42	Banco Colpatria	860034594-1
1	43	BANCO AGRARIO DE COLOMBIA o BANAGRARIO.	800037800-8
1	49	AV Villas	860035827-5
1	51	Banco Credifinanciera S.A.	900200960-9
1	52	Bancamía S.A.	900215071-1

Source: Superintendencia Financiera de Colombia

Last update: 2025-02-27

Table 2: Supervised banking establishments

Type	Code	Abbreviate Name	NIT
1	53	Banco W S.A.	900378212-2
1	54	Bancoomeva	900406150-5
1	55	Finandina Bic o Banco Finandina Bic o Finandina.	860051894-6
1	56	Banco Falabella S.A.	900047981-8
1	57	Banco Pichincha S.A.	890200756-7
1	58	Coopcentral	890203088-9
1	59	Banco Santander	900628110-3
1	60	Banco Mundo Mujer S.A.	900768933-8
1	62	Mibanco S.A.	860025971-5
1	63	Banco Serfinanza S.A.	860043186-6
1	64	Banco J.P. Morgan Colombia S.A., (la "Sociedad")	900114346-8
1	65	Lulo Bank S.A.	901383474-9
1	66	Banco BTG Pactual Colombia S.A.	901491551-0
1	67	BANCO UNIÓN S.A. (en adelante el "Banco" o la "Sociedad")	860006797-9
1	68	BANCO CONTACTAR S.A.	901702583-3

Source: Superintendencia Financiera de Colombia

Last update: 2025-02-27

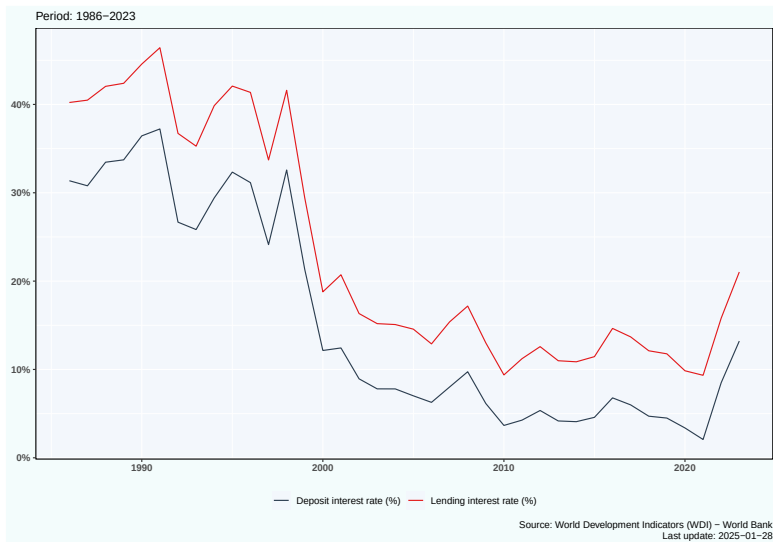


Figure 3: Deposit and lending interest rate in Colombia

Acknowledgments

- To my family that supports me
- To the taxpayers of Colombia and the **UMNG students** who pay my salary
- To the **Business Science** and **R4DS Online Learning** communities where I learn **R** and **π -thon**
- To the **R Core Team**, the creators of **RStudio IDE**, **Quarto** and the authors and maintainers of the packages **tidyverse**, **wbstats**, **tidyquant**, **ggrepel**, **lubridate**, **knitr**, **kableExtra**, **readxl**, **janitor** and **tinytex** for allowing me to access these tools without paying for a license
- To the **Linux kernel community** for allowing me the possibility to use some **Linux distributions** as my main **OS** without paying for a license



References I

- Cardenas, Mauricio. 2020. *Introducción a La Economía Colombiana*. 4th ed. Alfaomega.
- Colombia, Superintendencia Financiera de. 2021. "Circular Básica Contable y Financiera (Circular Externa 100 de 1995)." <https://www.superfinanciera.gov.co/jsp/loader.jsf?!Servicio=Publicaciones&ITipo=publicaciones&IFuncion=loadContenidoPublicacion&id=15466>.
- Freixas, Xavier, and Jean-Charles Rochet. 2008. *Microeconomics of Banking*. 2nd ed. Cambridge, Mass: MIT Press.